

Day 1 – Data Analytics Ability Assessment

SUPERSTORE SALES ANALYSIS & BUSINESS INSIGHTS

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Tools Used: **Python, Pandas, Matplotlib, Microsoft Power BI**

Dataset: **Superstore Sales Dataset (2011–2015)**

1. Introduction

The objective of this project is to analyze a sales dataset and identify meaningful business insights related to sales, profitability, order quantity, categories, regions, products, and time-based performance.

The analysis was carried out using **Python and Pandas** for data cleaning and exploratory data analysis, followed by **Microsoft Power BI** for creating an interactive business dashboard.

The main focus of the project was not only to visualize the data but also to understand what the data indicates from a business perspective and provide recommendations based on the findings.

2. Dataset Overview

The dataset used for this analysis is the **Superstore Sales Dataset covering the period from 2011 to 2015**.

The original dataset contains **51,290 records and 24 columns**. It includes information about customer orders, products, categories, regions, sales, quantities, discounts, profits, shipping costs, and order dates.

Some of the important fields used in the analysis are:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- City
- State
- Country
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity

- Discount
- Profit
- Shipping Cost
- Order Priority

These fields provide sufficient information to evaluate both **revenue performance and profitability**.

3. Data Cleaning and Preparation

Before performing the analysis, the dataset was inspected and cleaned using **Python and Pandas**.

The first step was to understand the structure of the dataset, including the number of records, columns, data types, and missing values.

The following cleaning and validation steps were performed:

~Missing Values

The dataset contained missing values in the **Postal Code** column. A total of **41,296 records** had missing postal codes.

Since postal code was not required for the main sales and profitability analysis, these missing values were retained rather than introducing potentially incorrect values.

~Duplicate Records

The dataset was checked for duplicate rows. **No duplicate records were identified**, so no rows were removed for duplication.

~Data Types

The date fields were converted into proper datetime format, while numerical fields such as Sales, Quantity, Discount, Profit, and Shipping Cost were converted into appropriate numeric data types.

~Date Validation

Order Date and Ship Date were checked to ensure that shipping did not occur before the corresponding order date.

No invalid shipping sequences were found.

The calculated shipping duration ranged from **0 to 7 days**, with an average shipping time of approximately **3.97 days**.

~Numerical Validation

Sales, Quantity, Discount, and Shipping Cost were checked for invalid values.

The analysis found:

- No negative Sales values
- No zero Sales values
- No negative or zero Quantity values

- No Discount values below 0
- No Discount values above 1
- No negative Shipping Cost values

After cleaning and preparation, the dataset contained **51,290 records** and **25 analytical columns**, including the calculated Shipping Days field.

```
import pandas as pd

df = pd.read_csv("/content/superstore_dataset2011-2015.csv", encoding="cp1252")

df = pd.read_csv("superstore_dataset2011-2015.csv", encoding="cp1252")

print("Rows:", len(df))
print("Columns:", len(df.columns))

df.head()
```

Rows: 51290
Columns: 24

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	...	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Shipping Cost	Order Priority	
0	42433	AG-2011-2040	1/1/2011	6/1/2011	Standard Class	TB-11280	Toby Braunhardt	Consumer	Constantine	Constantine	...	OFF-TEN-10000025	Office Supplies	Storage	Tenex Lockers, Blue	408.300	2	0.0	106.140	35.46	Medium
1	22253	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	JH-15985	Joseph Holt	Consumer	Wagga Wagga	New South Wales	...	OFF-SU-10000518	Office Supplies	Supplies	Acme Trimmer, High Speed	120.366	3	0.1	36.036	9.72	Medium
2	48883	HU-2011-1220	1/1/2011	5/1/2011	Second Class	AT-735	Annie Thurman	Consumer	Budapest	Budapest	...	OFF-TEN-10001585	Office Supplies	Storage	Tenex Box, Single Width	66.120	4	0.0	29.640	8.17	High
3	11731	IT-2011-3647632	1/1/2011	5/1/2011	Second Class	EM-14140	Eugene Maren	Home Office	Stockholm	Stockholm	...	OFF-PA-10001492	Office Supplies	Paper	Enemax Note Cards, Premium	44.865	3	0.5	-26.055	4.82	High
4	22255	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	JH-15985	Joseph Holt	Consumer	Wagga Wagga	New South Wales	...	FUR-FU-10003447	Furniture	Furnishings	Eldon Light Bulb, Duo Pack	113.670	5	0.1	37.770	4.70	Medium

5 rows x 24 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 24 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Row ID              51290 non-null  int64
1   Order ID            51290 non-null  object
2   Order Date          51290 non-null  object
3   Ship Date           51290 non-null  object
4   Ship Mode           51290 non-null  object
5   Customer ID         51290 non-null  object
6   Customer Name       51290 non-null  object
7   Segment             51290 non-null  object
8   City                51290 non-null  object
9   State               51290 non-null  object
10  Country             51290 non-null  object
11  Postal Code         9994 non-null   float64
12  Market              51290 non-null  object
13  Region              51290 non-null  object
14  Product ID          51290 non-null  object
15  Category            51290 non-null  object
16  Sub-Category        51290 non-null  object
17  Product Name        51290 non-null  object
18  Sales               51290 non-null  float64
19  Quantity            51290 non-null  int64
20  Discount            51290 non-null  float64
21  Profit              51290 non-null  float64
22  Shipping Cost       51290 non-null  float64
23  Order Priority       51290 non-null  object
dtypes: float64(5), int64(2), object(17)
memory usage: 9.4+ MB
```

```
df.isnull().sum()
```

```
***
0
Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID 0
Customer Name 0
Segment     0
City        0
State       0
Country     0
Postal Code 41296
Market      0
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
Quantity    0
Discount    0
Profit      0
Shipping Cost 0
Order Priority 0
dtype: int64
```

```

clean_df = clean_df.drop_duplicates()

clean_df["Shipping Days"] = (
    clean_df["Ship Date"] - clean_df["Order Date"]
).dt.days

print("Cleaning completed.")
print("Rows:", len(clean_df))
print("Columns:", len(clean_df.columns))

```

```

Cleaning completed.
Rows: 51290
Columns: 25

```

```

print("Rows after cleaning:", len(clean_df))
print("Duplicate rows:", clean_df.duplicated().sum())
print("Missing values:")
print(clean_df.isnull().sum())

```

```

*** Rows after cleaning: 51290
Duplicate rows: 0
Missing values:
Row ID          0
Order ID        0
Order Date      31223
Ship Date       31456
Ship Mode       0
Customer ID     0
Customer Name   0
Segment        0
City           0
State          0
Country        0
Postal Code     41296
Market         0
Region         0
Product ID     0
Category       0
Sub-Category   0
Product Name    0
Sales          0
Quantity       0
Discount       0
Profit         0
Shipping Cost   0
Order Priority  0
Shipping Days   37993
dtype: int64

```

```

print("Negative Sales:", (clean_df["Sales"] < 0).sum())
print("Invalid Quantity:", (clean_df["Quantity"] <= 0).sum())
print("Invalid Discount:", (
    (clean_df["Discount"] < 0) |
    (clean_df["Discount"] > 1)
).sum())

```

```

*** Negative Sales: 0
Invalid Quantity: 0
Invalid Discount: 0

```

```

print(
    "Invalid dates:",
    (
        clean_df["Order Date"].isna() |
        clean_df["Ship Date"].isna() |
        (clean_df["Ship Date"] < clean_df["Order Date"])
    ).sum()
)

```

```
Invalid dates: 37993
```

```
clean_df.shape
```

```
*** (51290, 25)
```

```

cleaned_file = "superstore_cleaned.csv"

clean_df.to_csv(cleaned_file, index=False)

print("Cleaned dataset exported successfully:", cleaned_file)

Cleaned dataset exported successfully: superstore_cleaned.csv

```

4. Exploratory Data Analysis

After cleaning the data, exploratory analysis was performed to understand the overall business performance and identify important patterns.

The analysis focused on:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Profit Margin
- Sales trends over time
- Category performance
- Regional performance
- Sub-category performance
- Product-level performance
- Discount and profitability relationship

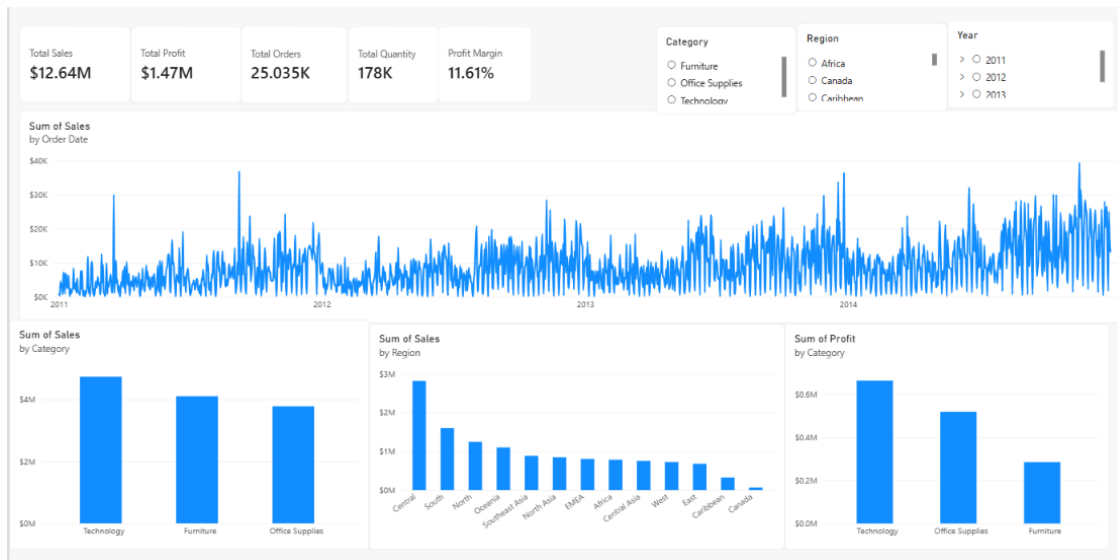
4.1 Key Performance Indicators

The overall performance of the dataset is summarized below:

KPI	Value
Total Sales	\$12,642,501.91
Total Profit	\$1,467,457.29
Total Orders	25,035
Total Quantity	178,312
Profit Margin	11.61%
Average Shipping Time	3.97 days

These KPIs provide a high-level understanding of the business performance represented by the dataset.

The total sales of approximately **\$12.64 million** generated approximately **\$1.47 million in profit**, resulting in an overall profit margin of **11.61%**.



4.2 Sales Trend Analysis

Sales were analyzed across the available order dates to understand how sales activity changed over time.

The sales trend indicates an overall increase in sales activity toward the later part of the dataset period, although individual dates show considerable fluctuations.

This indicates that sales performance was not completely consistent throughout the period and that certain periods contributed significantly more to overall revenue.



4.3 Category-wise Sales and Profit

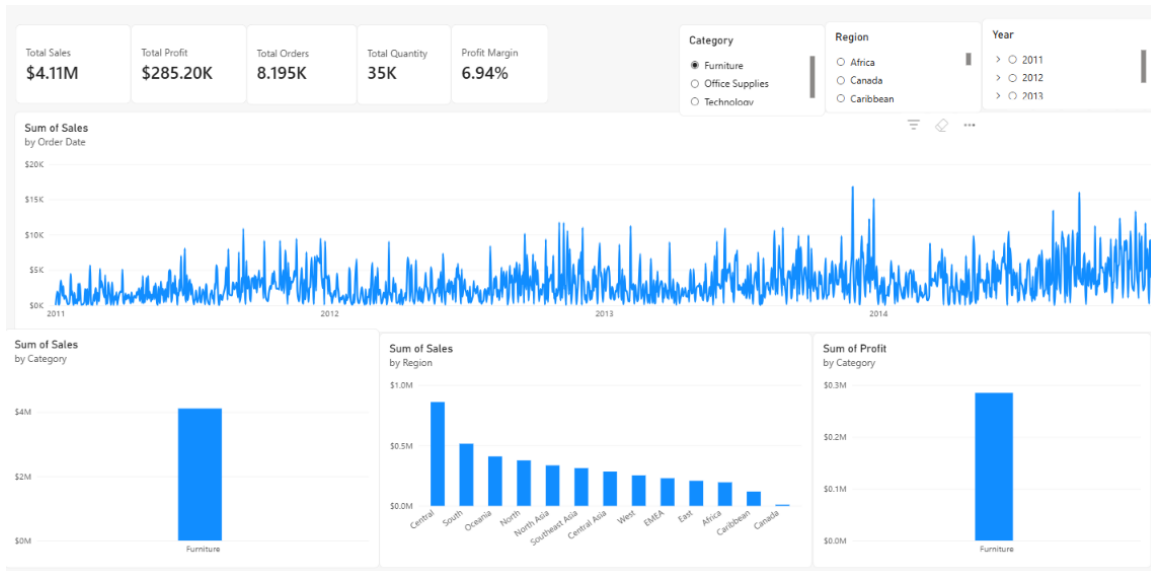
The dataset contains three major product categories:

- Technology
- Furniture
- Office Supplies

Among these categories, **Technology is the strongest-performing category**, contributing the highest overall sales and profit.

This indicates that Technology products play an important role in the overall business performance.

The comparison between sales and profit is particularly useful because a category can generate high sales without necessarily generating the highest profit.

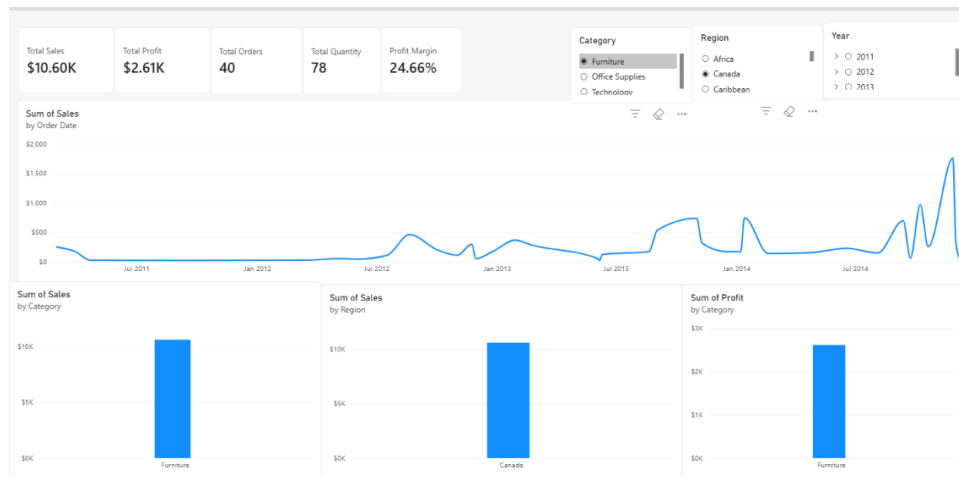


4.4 Region-wise Performance

Regional analysis was performed to understand how different geographical regions contribute to overall sales.

The **Central region records the highest sales** among the regions in the dataset.

This suggests that the Central region represents an important market and could provide useful information about successful products, customer segments, and sales strategies.



5. Product Performance Analysis

Product-level analysis was performed to identify products that contribute strongly to revenue as well as products that generate losses.

Top Products by Sales

Product	Sales	Profit
Apple Smart Phone, Full Size	\$86,935.78	\$5,921.58
Cisco Smart Phone, Full Size	\$76,441.53	\$17,238.52

Motorola Smart Phone, Full Size	\$73,156.30	\$17,027.11
Nokia Smart Phone, Full Size	\$71,904.56	\$9,938.20
Canon imageCLASS 2200 Advanced Copier	\$61,599.82	\$25,199.93

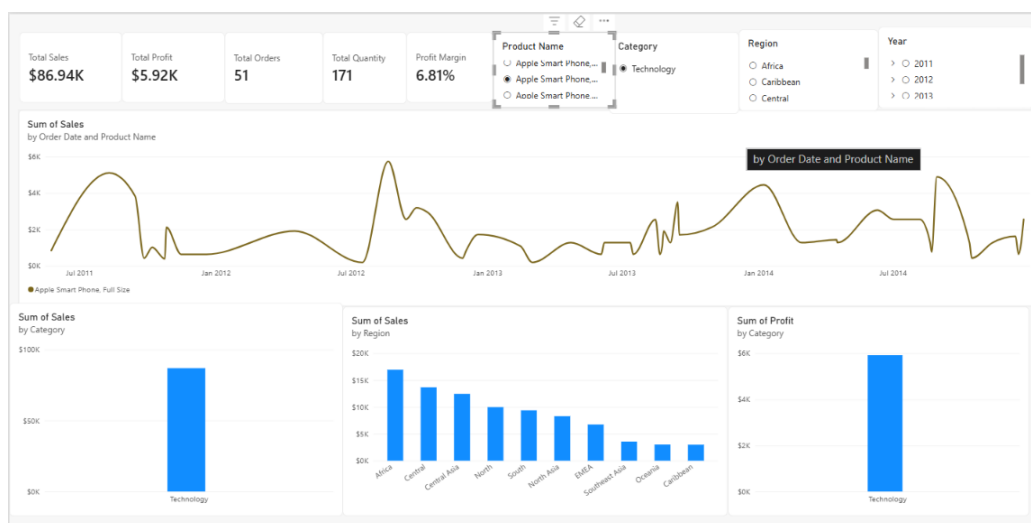
These products demonstrate strong revenue contribution, although their profitability varies considerably.

Loss-Making Products

The analysis also identified several products with significant negative profit:

Product	Sales	Profit
Cubify CubeX 3D Printer Double Head Print	\$11,099.96	-\$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	\$16,829.90	-\$4,589.97
Motorola Smart Phone, Cordless	\$38,931.04	-\$4,447.04
Cubify CubeX 3D Printer Triple Head Print	\$7,999.98	-\$3,839.99
Bevis Round Table, Adjustable Height	\$5,654.80	-\$3,649.89

This shows that product performance should not be evaluated using sales alone. Profitability must also be considered when making business decisions.



6. Discount and Profitability Analysis

Discount levels were analyzed to understand their relationship with profitability.

There were **22,281 records with a discount greater than zero**. These discounted transactions generated approximately **\$5.65 million in sales**, but collectively recorded a profit of approximately **-\$303,237.98**.

The average profit also decreases significantly at higher discount levels.

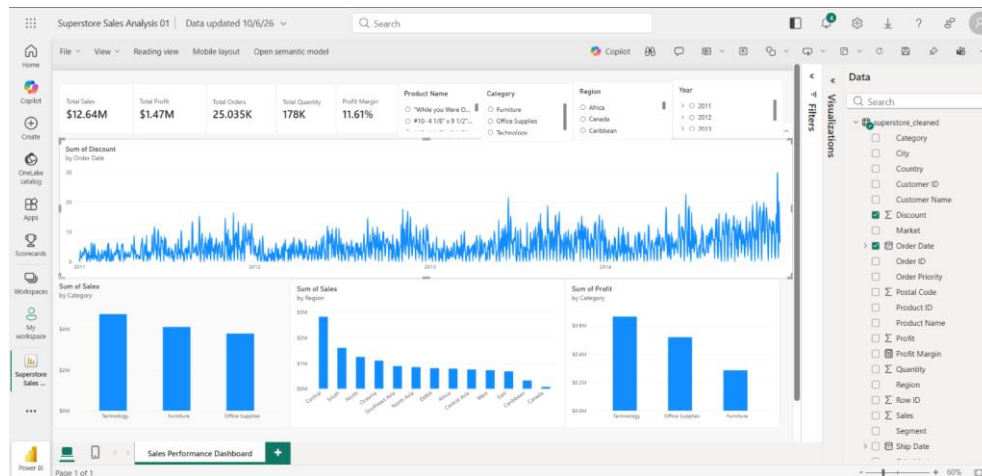
For example:

Discount	Average Profit
0%	\$61.04
10%	\$63.68
20%	\$23.55

30%	-\$57.90
40%	-\$45.25
50%	-\$97.14
85%	-\$1,534.33

The results indicate a strong negative association between higher discount levels and profitability.

This does not necessarily prove that discounts directly cause losses, but it highlights the need for careful discount management.



7. Business Insights

Based on the analysis, the following key business insights were identified.

~Technology is the strongest category

Technology is the strongest overall category in the dataset in terms of both sales and profit. It generates approximately **\$4.7 million in sales** and approximately **\$0.69 million in profit**.

This indicates that Technology is a major contributor to overall business performance.

~Central is the leading region by sales

The **Central region** records the highest sales among the analyzed regions.

The business can further investigate what products, customers, or sales strategies are contributing to the strong performance of this region and determine whether similar strategies can be applied elsewhere.

~High sales do not always mean high profitability

Phones generate the highest sub-category sales at approximately **\$1.71 million**, while Copiers generate the highest sub-category profit at approximately **\$258.6K**.

This demonstrates that sales volume alone is not sufficient for evaluating business performance. Profitability must also be considered.

~Tables are a major profitability concern

The Tables sub-category generated approximately **\$757K in sales** but recorded a **loss of \$64,083.39**.

This is a significant profitability issue and suggests that the pricing, discounting, product costs, or other operational factors associated with Tables should be reviewed.

~Higher discounts are associated with lower profitability

Discounted transactions generated approximately **\$5.65 million in sales but recorded a combined loss of approximately \$303K**.

The average profit becomes increasingly negative at higher discount levels, with an average loss of approximately **\$1,534 per record at an 85% discount**.

This indicates that aggressive discounting can significantly reduce profitability.

~Several individual products are consistently loss-making

The **Cubify CubeX 3D Printer Double Head Print** generated approximately **\$11.1K in sales but a loss of \$8.88K**, making it the largest loss-making product identified in the analysis.

Other products also show significant negative profitability despite generating sales.

These products should be reviewed individually before deciding whether they should continue to be sold under the current pricing and cost structure.

8. Power BI Dashboard

An interactive dashboard was developed using **Microsoft Power BI** to present the key findings in a business-friendly format.

The dashboard includes:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Profit Margin
- Sales trend
- Sales by Category
- Sales by Region
- Profit by Category
- Year filter
- Region filter
- Category filter

The interactive filters allow users to explore the dashboard based on different years, regions, and product categories.

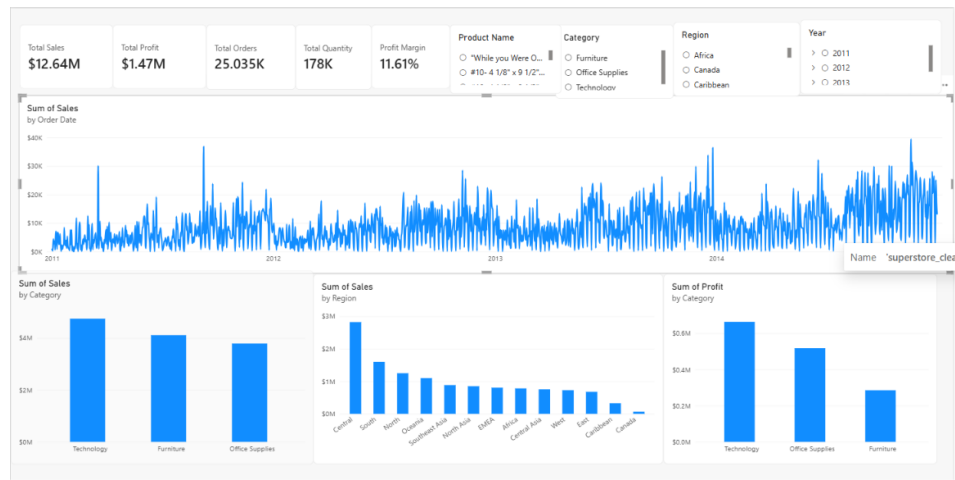


Figure : Superstore Sales Performance Dashboard

9. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Microsoft Power BI

10. Conclusion

The Superstore sales analysis successfully transformed the raw transactional dataset into a structured analysis and interactive business dashboard.

The analysis identified several important patterns. **Technology emerged as a strong contributor to both sales and profit**, while **Central recorded the highest regional sales**. At the same time, the analysis identified profitability concerns in areas such as **Tables, loss-making products, and heavily discounted transactions**.

The findings demonstrate that sales volume alone is not sufficient to evaluate business performance. A more effective approach is to consider **sales, profit, profit margin, discount levels, product performance, and regional performance together**.

The Power BI dashboard provides an interactive way to explore these findings and supports more informed, data-driven business decisions.

Deliverables

- **Raw Dataset:** Superstore Sales Dataset
- **Cleaned Dataset:** superstore_cleaned.csv
- **Analysis Notebook:** sales_analysis_cleaning.ipynb
- **Power BI Dashboard:** Superstore Sales Analysis Dashboard

- **Business Insights:** Included in this report
- **Business Recommendations:** Included in this report
- **Dashboard Screenshot:** Included in this report
- **Explanation Video:** Submitted separately through Google Drive